Name: _____

WORKSHEET

Factorial notation

1 Expand the following.

a 4! _____

b 7! _____

c 13! _____

2 Calculate the following, using technology where necessary.

a 8! _____

b 11! _____

c 13! _____

d 22! _____

e $\frac{78!}{73!}$ _____

f $\frac{100!}{95!}$ _____

3 Calculate the following without using technology.

a 4! _____

b 6! _____

c $\frac{10!}{7!}$ _____

d $\frac{11!}{9!}$ _____

e $\frac{20!}{18!}$ _____

f $\frac{7!}{4!}$ _____

4 Find: **i** n and **ii** $(n-3)!$ If $n!$ is equal to:

a 120

b 5040

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c 362 880

d 479 001 600

5 Solve the following problems.

a A lucky dip contains 6 different prizes. In how many different ways can six children win these prizes?

b If a phone number has 8 digits, how many different phone numbers exist that do not repeat any numbers?

c A committee of 6 needs to be chosen from a pool of 20 people. In how many ways is this possible?

d A case-sensitive password has 5 letters or numbers. How many different combinations are possible if none are repeated?

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Answers

1 **a** $4 \times 3 \times 2 \times 1$

b $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

c $13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

2 **a** 40 320

b 39 916 800

c 6 227 020 800

d 1 124 000 727 777 607 680 000

e 2 533 330 800

f 9 034 502 400

3 **a** 24

b 720

c 720

d 110

e 380

f 210

4 **a** **i** $n = 5$ **ii** $(n - 3)! = 2$

b **i** $n = 7$ **ii** $(n - 3)! = 24$

c **i** $n = 9$ **ii** $(n - 3)! = 720$

d **i** $n = 10$ **ii** $(n - 3)! = 3\,628\,800$

5 **a** 720

b 40 320

c 27 907 200

d 45 239 040

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